

Name:			Date:	
Teacher:	<u>Solutions</u>			
 Baldivis Secondary College	Year 9 Mathematics Topic – Non-Linear Equations		SCORE: / 28	
	<u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	30 minutes			
Weighting:	8 %			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.			

1. Consider the following two equations: $y=2x^2$ and $y=2^x$ [2, 6, 1, 2 - 11 marks]

a. Complete the table below using the equation $y=2x^2$

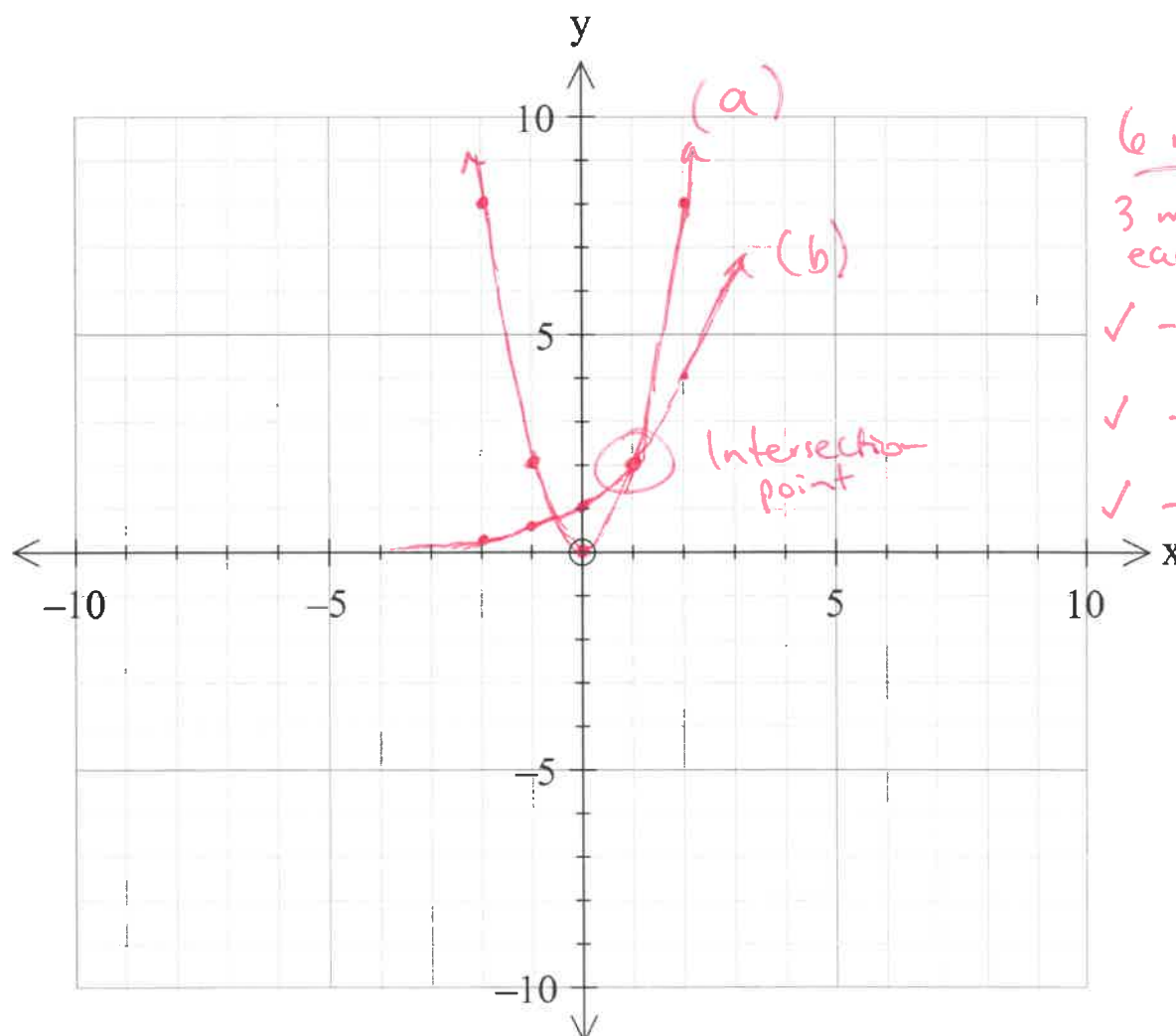
x	-2	-1	0	1	2
y	8	2	0	2	8

2 mks:
1 mk - 3 correct
1 mk - all 5 correct

b. For the equation $y=2^x$ we are given the completed table of values

x	-2	-1	0	1	2
y	0.25	0.5	1	2	4

c. Plot these two graphs on the cartesian plane below



6 mks total
3 mks for each graph
✓ - 3 pts plotted
✓ - all 5 pts plotted
✓ - smooth curves

d. Determine **one** of the intercept points of the two graphs

$(1, 2)$ ✓ - 1mk

(the other intercept too difficult)

e. Describe what type of graph each table forms

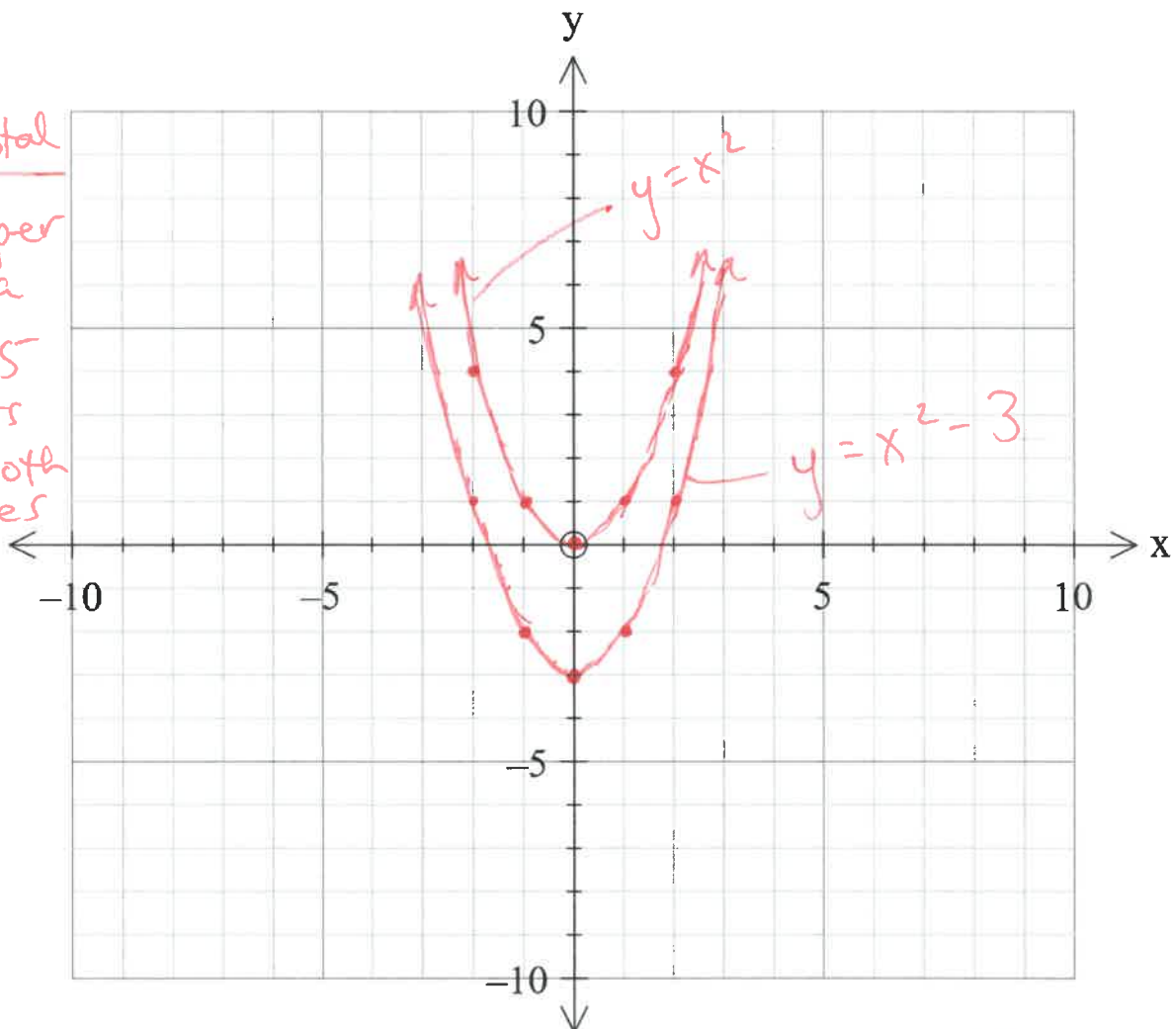
a) Quadratic / parabola ✓ - 1mk
b) Exponential ✓ - 1mk

2. The equations $y=x^2$ and $y=x^2-3$ have been put into a table of values below. Using the axis provided, construct the graphs and then describe the difference between the two graphs.

[4, 1 - 5 marks]

x =	-2	-1	0	1	2
$y=x^2$	4	1	0	1	4
$y=x^2-3$	1	-2	-3	-2	1

4 mks total
2 mks per parabola
✓ - all 5 points
✓ - smooth curves



Describe the difference:

✓ $y = x^2 - 3$ is shifted (translated) 3 units down
1 mk -

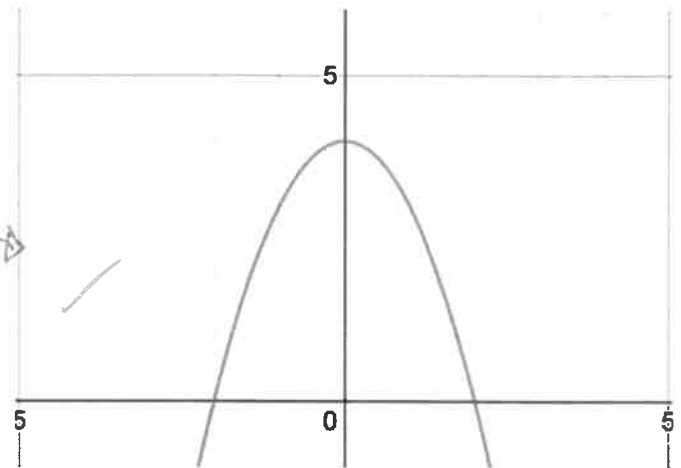
3. Given the following tables, match the rules to the graphs

[3 Marks]

a.

x	-2	0	2
y	2	0	-8

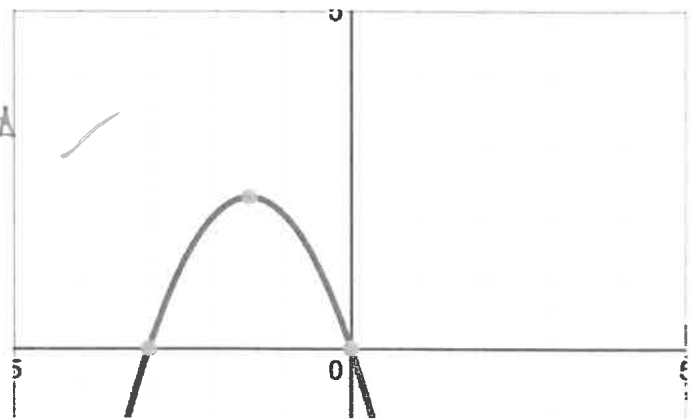
i.



b.

x	-2	0	2
y	0	0	10

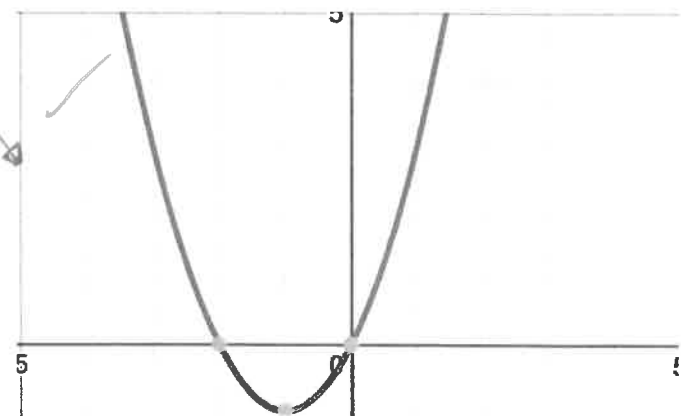
ii.



c.

x	-2	0	2
y	0	4	0

iii.

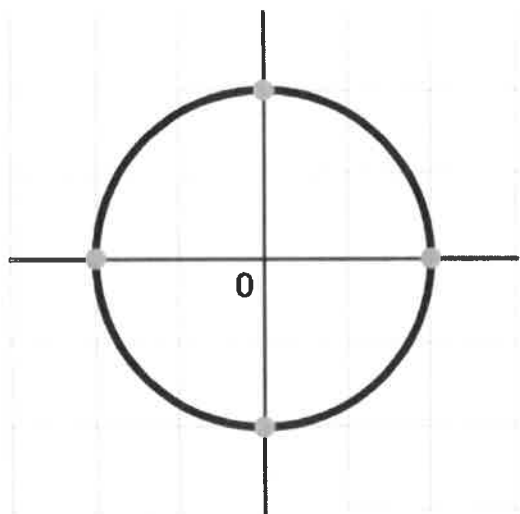


a → ii
b → iii
c → i

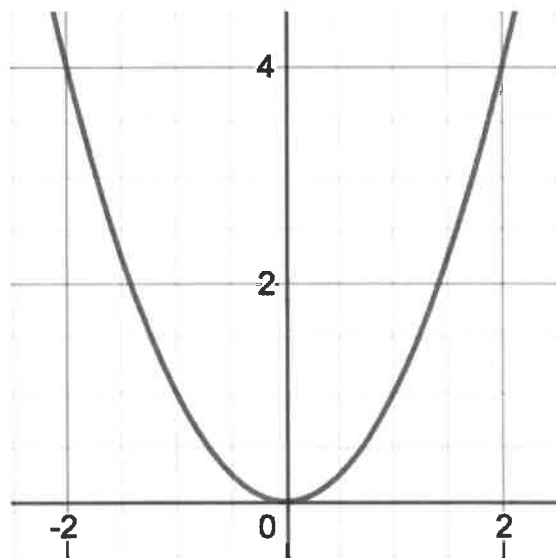
4. Write the rule below underneath their correct graphs

[4 Marks]

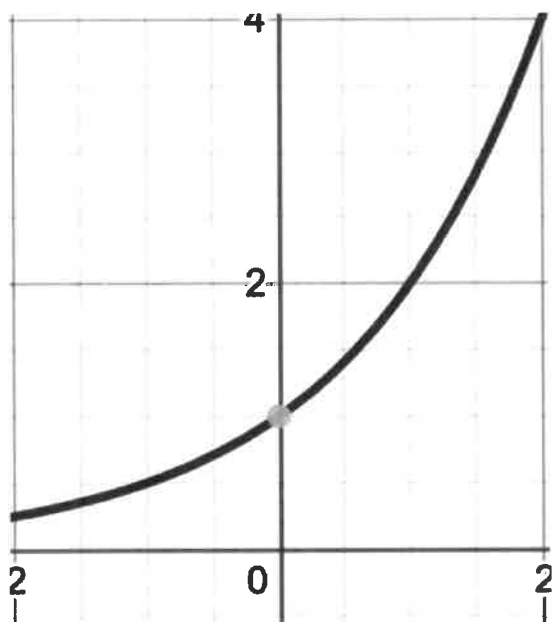
- a. $y = 2x$
- b. $y = x^2$
- c. $y = 2^x$
- d. $y^2 + x^2 = 4$



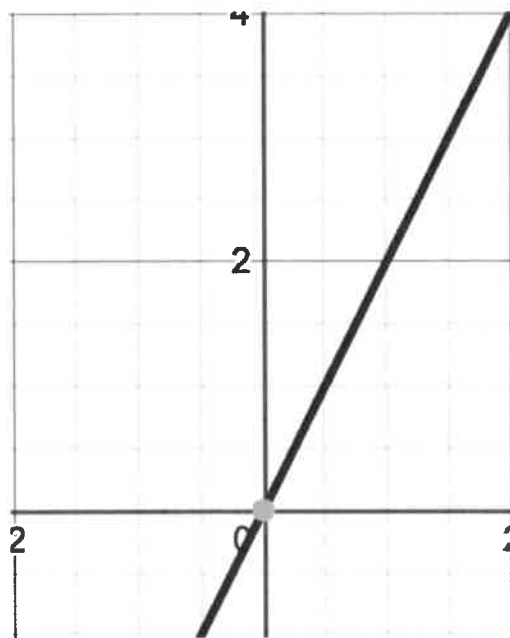
$y^2 + x^2 = 4$ ✓



$y = x^2$ ✓



$y = 2^x$ ✓



$y = 2x$ ✓

5. Kalgoorlie has one of the largest open cut mines in the Southern hemisphere. If you tossed a rock into it, it would form an arc with the equation of $h = -s^2 + 2s + 2$, where h is the height in metres, and s is the time in seconds. [2, 3 - 5 marks]

a. Determine the height of the rock as it is released

substitute $s = 0$

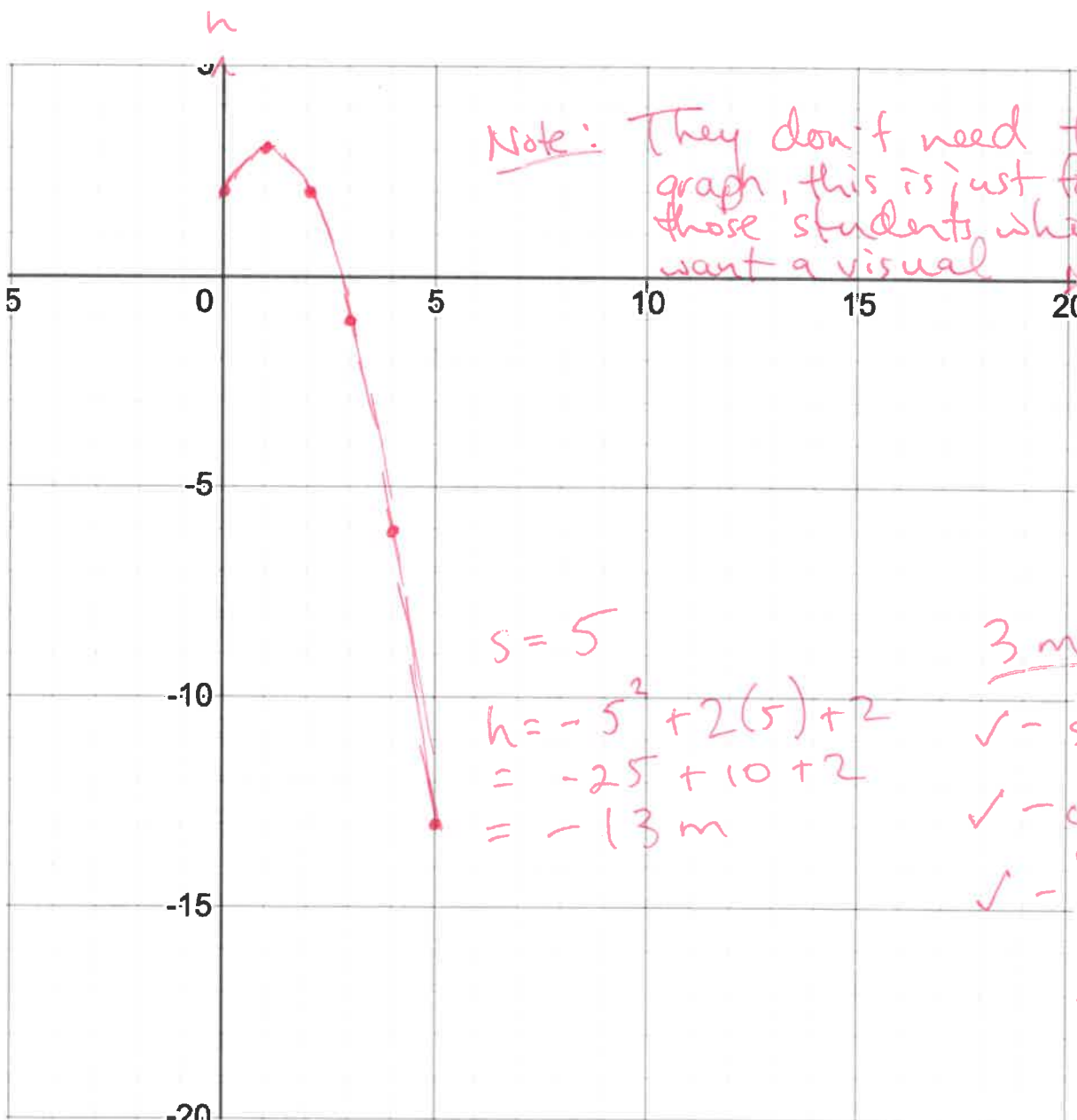
$$h = -(0)^2 + 2(0) + 2$$

$$= 2 \text{ meters}$$

1 mk: sub in 0

1 mk: 2 m answer

b. Determine the height of the rock after 5 seconds



End of Test

